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Is More Information Better? The Effects of “Report Cards” on Health Care Providers

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1 Big picture

- **Question.** Do public quality report cards improve patient welfare in health care markets, or do they distort provider behavior by inducing patient selection?
- **Setting.** The paper studies coronary artery bypass graft (CABG) surgery report cards in New York and Pennsylvania during the early 1990s.
- **Policy.** New York began public reporting of hospital CABG mortality in 1990 and surgeon-specific mortality in 1992. Pennsylvania began publishing hospital- and surgeon-specific risk-adjusted CABG mortality in 1992.
- **Core tension.** Report cards may reduce informational asymmetries and help patients find better hospitals, but they also give hospitals and physicians incentives to avoid high-risk patients whose outcomes could hurt published rankings.
- **Main empirical design.** The paper compares changes over time in New York and Pennsylvania with changes in other states, and also uses neighbouring states as a robustness comparison group.
- **Main population.** The key population is Medicare beneficiaries admitted with acute myocardial infarction (AMI), not only patients who received CABG.
- **Why AMI patients matter.** AMI is a population at risk for intensive cardiac treatment. Studying all AMI patients avoids the selection problem that arises when one studies only patients who actually receive CABG.
- **Main result.** Report cards induced provider selection and changed treatment patterns. They led to fewer severely ill CABG patients at nonteaching hospitals and more sorting of sick patients into teaching hospitals.
- **Treatment result.** Report cards increased CABG use for healthier AMI patients but delayed intensive cardiac treatment and reduced some alternative intensive treatments such as PTCA.
- **Welfare result.** Report cards increased hospital expenditures and worsened some health outcomes, including readmission with heart failure and mortality among sicker patients.
- **Takeaway.** Public information can backfire when the measured performance metric is risk-adjusted imperfectly and providers can respond by selecting patients or delaying treatment.

2 Incremental contribution of the paper

2.1 Contribution relative to the quality-report-card literature

- Earlier studies of New York CABG report cards often focused only on patients receiving CABG and interpreted lower observed CABG mortality as evidence of quality improvement.
- Dranove et al. show that this approach can be misleading because the treated CABG population is itself an outcome of the policy.
- By examining the full AMI population at risk for CABG, the paper evaluates whether report cards improve welfare for all potentially affected patients, not only for the selected group that receives surgery.

Interpretation: The key methodological contribution is to avoid conditioning on a post-treatment variable. CABG receipt can be changed by the report cards, so focusing only on CABG recipients can mistake selection for quality improvement.

2.2 Contribution relative to adverse-selection and information-disclosure theory

- The paper provides empirical evidence that more public information is not always welfare-improving.
- It shows that information disclosure can change provider incentives in ways not captured by simple models in which better information improves matching.
- Report cards provide useful quality information but also create strong incentives to avoid patients whose outcomes are hard to risk-adjust.

Interpretation: The paper is an applied test of the idea that performance measurement can distort behavior when agents control the composition of cases being measured.

2.3 Contribution relative to health-care policy evaluation

- The paper separates several channels: incidence effects, matching effects, quantity effects, treatment-delay effects, cost effects, and mortality/readmission effects.
- It evaluates social welfare by comparing resource use and patient outcomes rather than only utilization or reported quality scores.
- It shows that even if report cards improve matching of sicker patients to better hospitals, this can be offset by patient selection and treatment delays.

Interpretation: The paper's contribution is not just showing that report cards changed behavior. It shows that the full equilibrium response lowered patient and social welfare in the short run.

2.4 Contribution relative to empirical industrial organization and regulation

- The paper treats provider behavior as endogenous to regulation.
- It demonstrates that mandated disclosure changes both demand-side matching and supply-side selection incentives.
- The evidence is useful beyond health care: any report-card policy in education, finance, or labor markets can face the same selection problem.

Interpretation: The broader lesson is that regulatory information design must anticipate how measured agents will change what they do and whom they serve.

3 Institutional background

3.1 CABG report cards

- CABG surgery is a major cardiac procedure used to treat severe coronary artery disease and some AMI patients.
- New York and Pennsylvania publicly reported CABG mortality outcomes at the hospital and surgeon level.
- The report cards were risk-adjusted, but risk adjustment was necessarily incomplete because providers observe clinical information that data analysts do not observe.
- Report cards therefore created incentives both to improve quality and to avoid patients likely to worsen mortality statistics.

Interpretation: Report cards attempt to solve an asymmetric-information problem between providers and patients, but they also create a new agency problem between providers and report-card designers.

3.2 Why selection is plausible

- CABG is often elective or semi-elective, so providers can influence whether and when a patient receives it.
- Sick patients may have more to gain from high-quality providers, but they are also more likely to die and therefore more likely to harm a provider's report-card ranking.
- Low-quality providers may especially prefer to avoid sick patients because the outcome gap between high- and low-quality providers is likely largest for high-risk patients.
- Risk-averse providers may avoid severe cases if risk adjustment does not fully compensate them for outcome variance.

Interpretation: The key empirical prediction is that after report cards, the pool of CABG patients should become healthier, especially at nonteaching or lower-quality hospitals.

4 Data and measurement

4.1 Medicare claims data

- The main data are longitudinal Medicare claims for elderly beneficiaries admitted with AMI or receiving CABG between 1987 and 1994.
- The data allow the authors to observe prior-year hospital spending and days in hospital, treatment receipt, timing of treatment, readmissions, and mortality.
- The AMI cohort is the primary at-risk population because AMI generally leads to hospital admission and is less likely to be selected by providers.
- The CABG cohort is used for comparison and to replicate prior CABG-only analyses.

Interpretation: The Medicare data let the authors measure both pre-treatment patient severity and post-treatment outcomes for a national elderly population.

4.2 Hospital data

- Hospital characteristics come from the American Hospital Association survey.
- Hospital controls include rural location, size, ownership status, teaching status, and state-level hospital counts.
- Teaching status is especially important because teaching hospitals are treated as higher-quality or more specialized cardiac-care providers.

Interpretation: The hospital data enable tests of whether report cards shifted high-risk patients toward teaching hospitals.

4.3 Patient severity measures

- Patient severity before admission is measured using total hospital expenditures in the year prior to admission.
- A second severity measure is total days in hospital in the year prior to admission.
- A prior hospital admission is used as a binary proxy for being sicker.
- Hospital-level matching analysis uses within-hospital coefficients of variation and means of these severity measures.

Interpretation: Prior utilization is a practical claims-based severity proxy. It captures hidden health status that may not be fully observed by report-card analysts.

4.4 Treatment and outcome measures

- Intensive treatments include CABG, percutaneous transluminal coronary angioplasty (PTCA), and cardiac catheterization.
- The paper measures treatment within one year of AMI admission and treatment within one day of admission.
- Cost is measured by total hospital expenditures in the year after AMI admission.
- Health outcomes include readmission with AMI, readmission with heart failure, and all-cause mortality within one year.

Interpretation: The timing measures are important because delaying treatment can harm patients even if the total one-year treatment probability changes little.

5 Empirical specifications

5.1 Hospital-level incidence and matching model

The hospital-level model is

$$\ln(h_{lst}) = A_s + B_t + \gamma Z_{lst} + \pi L_{st} + \theta N_{st} + \varepsilon_{lst}. \quad (1)$$

- h_{lst} is a hospital-level health-status measure for hospital l in state s and year t .
- A_s are state fixed effects.
- B_t are year fixed effects.
- Z_{lst} are hospital controls.
- L_{st} indicates report-card treatment in New York and Pennsylvania after the relevant effective dates.
- N_{st} controls flexibly for the number of hospitals in the state.

Interpretation: When h_{lst} is mean severity for CABG patients, a negative π implies that report cards made the CABG patient pool healthier. When h_{lst} is the coefficient of variation, a negative π implies more within-hospital homogeneity and thus more sorting.

5.2 Teaching-hospital matching test

To test whether sick patients are redirected to teaching hospitals, the paper estimates equation (1) with an interaction:

$$\ln(h_{lst}) = A_s + B_t + \gamma Z_{lst} + \pi L_{st} + \rho(L_{st} \times Teaching_l) + \theta N_{st} + \varepsilon_{lst}. \quad (1)$$

Interpretation: A positive ρ means that report cards increase severity at teaching hospitals relative to nonteaching hospitals. This is evidence of improved matching to high-quality or high-capacity providers.

5.3 Patient-level quantity model

For treatment receipt, the paper estimates

$$C_{kst} = A_s + B_t + \gamma Z_{kst} + \pi L_{st} + \varepsilon_{kst}. \quad (2)$$

- C_{kst} equals one if patient k in state s and year t receives CABG within one year of AMI admission.
- The same equation is estimated for PTCA and catheterization.
- Z_{kst} includes rural residence, gender, race, age categories, and interactions among demographic variables.

Interpretation: A positive π implies that report cards increase the probability that an AMI patient receives the treatment.

5.4 Patient-level incidence and welfare model

To test differential effects for sick versus healthy patients, the paper estimates

$$Y_{kst} = A_s + B_t + \gamma Z_{kst} + \pi L_{st} + \theta w_{kst} + \rho(L_{st} \times w_{kst}) + \varepsilon_{kst}. \quad (3)$$

- Y_{kst} can be treatment receipt, hospital expenditures, readmission, or mortality.
- w_{kst} is a patient severity measure, such as prior-year admission.
- The interaction $L_{st} \times w_{kst}$ captures whether report cards have different effects for sick patients.

Interpretation: This equation is central for welfare analysis. If report cards increase costs and worsen outcomes, the net welfare effect is negative.

6 Identification and empirical design

6.1 Treatment and comparison groups

- New York and Pennsylvania are treatment states because they adopted mandatory CABG report cards.
- The main comparison group is all other states.
- A robustness comparison uses nearby states: Connecticut, Maryland, and New Jersey.
- The paper considers two effective-date assumptions: New York in 1991 and Pennsylvania in 1993, or both effective in 1993.

Interpretation: The difference-in-differences design compares changes in treatment states with changes elsewhere around the report-card introduction.

6.2 Key identification assumptions

- AMI admission composition is assumed not to be directly affected by CABG report cards, especially in the short run.
- Other states provide a reasonable counterfactual for trends in treatment, severity, expenditures, and outcomes.
- Patient severity proxies capture relevant differential risk that providers may observe but report-card analysts may not fully adjust for.

Interpretation: The AMI population is critical because it reduces selection bias. AMI is an emergency admission, while CABG receipt is partly elective and can respond to provider incentives.

6.3 Potential threats

- New York and Pennsylvania may have had different pre-existing trends in cardiac care.
- Other report-card or discharge-abstract reporting systems in other states could contaminate controls.
- Claims-based severity proxies are imperfect.
- Providers may respond differently over the long run than over the short run studied here.

Interpretation: The paper addresses these concerns with alternative timing assumptions, neighbouring-state comparisons, hospital-level and patient-level models, and CABG-only replications.

7 Tables and figures

7.1 Table 1: severity trends before and after report cards

Read:

- Table 1 compares mean prior-year hospital expenditures in 1990 and 1994.
- For all AMI patients, prior-year expenditures rose similarly in New York/Pennsylvania and in comparison states.
- For all CABG patients, prior-year expenditures fell in New York/Pennsylvania but did not fall similarly elsewhere.
- For AMI patients who received CABG, prior-year expenditures also fell in New York/Pennsylvania.

Economic message: This preview suggests patient selection: after report cards, CABG recipients in the report-card states became healthier relative to CABG recipients elsewhere, while the overall AMI population did not.

	1990 (1)	1994 (2)	Percentage Change (3)
A. All AMI Patients			
N.Y. and Pa.	\$3,110	\$3,373	.0846
All other states	2,660	2,910	.0940
Conn., Md., and N.J. only	3,055	3,318	.0861
B. All Patients Receiving CABG within One Year of Admission			
N.Y. and Pa.	4,850	4,511	-.0699
All other states	3,657	3,660	.0008
Conn., Md., and N.J. only	5,015	4,934	-.0162
C. AMI Patients Receiving CABG within One Year of Admission			
N.Y. and Pa.	1,867	1,702	-.0883
All other states	1,537	1,585	.0312
Conn., Md., and N.J. only	1,911	1,859	-.0272

7.2 Tables 2 and 3: hospital-level selection and matching

Read:

- Table 2 shows that report cards reduced the mean severity of CABG patients and reduced within-hospital variation in CABG patient severity.
- These effects are not present for all AMI patients, supporting the claim that the CABG patient pool changed.
- Table 3 shows that after report cards, teaching hospitals in New York and Pennsylvania received more severe CABG patients relative to nonteaching hospitals.

Economic message: Report cards changed both incidence and matching. Providers treated fewer sick patients overall in CABG, but the sick patients who did receive CABG were more likely to be treated at teaching hospitals.

TABLE 2
EFFECTS OF REPORT CARDS ON THE WITHIN-HOSPITAL COEFFICIENT OF VARIATION AND MEAN OF PATIENTS' HEALTH STATUS BEFORE TREATMENT: MEDICARE BENEFICIARIES WITH AMI AND MEDICARE BENEFICIARIES RECEIVING CABG, 1987-94

DEPENDENT VARIABLE	BENEFICIARIES RECEIVING CABG		BENEFICIARIES WITH AMI	
	Assumes Report Cards Effective 1991 in N.Y. and 1993 in Pa. (1)	Assumes Report Cards Effective 1993 in N.Y. and Pa. (2)	Assumes Report Cards Effective 1991 in N.Y. and 1993 in Pa. (3)	Assumes Report Cards Effective 1993 in N.Y. and Pa. (4)
ln(mean of patients' total hospital expenditures one year prior to admission)	-3.92** (1.52)	-5.30** (1.10)	3.37** (1.32)	1.55 (2.25)
ln(mean of patients' total days in hospital one year prior to admission)	-3.74** (1.84)	-4.51** (1.54)	1.11 (2.76)	1.56 (2.95)
ln(CV of patients' total hospital expenditures one year prior to admission)	3.60** (1.39)	3.60** (1.77)	-2.32** (.64)	-2.43** (.66)
ln(CV of patients' total days in hospital one year prior to admission)	.94 (2.22)	2.74 (3.53)	-4.79** (1.79)	-4.98** (2.01)

NOTE.—Each table entry represents a separate model. Standard errors are based on an estimator of the variance-covariance matrix that is consistent in the presence of heteroscedasticity and of any correlation of regression errors within states over time. Coefficients and standard errors are multiplied by 100 to facilitate interpretation. Each observation is weighted by the number of patients admitted to the hospital in the cohort in question. Sample sizes for AMI patients, coefficient of variation of expenditures, 37,672; coefficient of variation of length of stay, 37,681; mean expenditures, 38,686; mean of length of stay, 38,084. Regressions also include controls for number of hospitals in state of residence.

Table 2: selection and within-hospital sorting

TABLE 3
EFFECTS OF REPORT CARDS FOR TEACHING AND ALL OTHER HOSPITALS ON THE MEAN OF PATIENTS' HEALTH STATUS BEFORE TREATMENT: MEDICARE BENEFICIARIES WITH AMI AND MEDICARE BENEFICIARIES RECEIVING CABG, 1987-94

DEPENDENT VARIABLE	BENEFICIARIES RECEIVING CABG		BENEFICIARIES WITH AMI	
	Report Cards Effective 1991 in N.Y. and 1993 in Pa. (1)	(Report Card Effective 1991 in N.Y. and 1993 in Pa.) × Teaching Hospital (2)	Report Cards Effective 1991 in N.Y. and 1993 in Pa. (3)	(Report Card Effective 1991 in N.Y. and 1993 in Pa.) × Teaching Hospital (4)
ln(mean of patients' total hospital expenditures one year prior to admission)	-18.63** (2.42)	19.78** (2.20)	-1.78 (3.95)	15.05** (7.46)
ln(mean of patients' total days in hospital one year prior to admission)	-11.38** (3.03)	10.28** (1.70)	-2.06 (4.89)	9.27 (6.11)

NOTE.—Each table entry represents a separate model. Standard errors are based on an estimator of the variance-covariance matrix that is consistent in the presence of heteroscedasticity and of any correlation of regression errors within states over time. Coefficients and standard errors are multiplied by 100 to facilitate interpretation. Each observation is weighted by the number of patients admitted to the hospital in the cohort in question. Sample size for coefficient of variation of expenditures, 37,672; for coefficient of variation of length of stay, 37,681; for mean expenditures, 38,686; for mean of length of stay, 38,084. Regressions also include controls for number of hospitals in state of residence.

** Significantly different from zero at the 5 percent level.

Table 3: matching to teaching hospitals

7.3 Table 4: treatment rates for AMI patients

Read:

- Table 4 studies CABG, PTCA, and catheterization among AMI patients.
- Report cards increase one-year CABG for healthier patients.
- Report cards reduce one-day CABG, PTCA, and catheterization rates, indicating treatment delays.
- PTCA falls, particularly for sicker patients.

Economic message: Report cards did not simply redirect patients to better care. They changed treatment timing and substitution across intensive cardiac procedures.

TABLE 4
EFFECTS OF REPORT CARDS ON CABG, PTCA, AND CATHETERIZATION RATES: MEDICARE BENEFICIARIES WITH AMI, 1987-94

DEPENDENT VARIABLE	A. ASSUMES REPORT CARDS EFFECTIVE 1991 IN N.Y. AND 1993 IN PA.			B. ASSUMES REPORT CARDS EFFECTIVE 1993 IN N.Y. AND PA.		
	Effect of Report Cards (1)	Admission to Hospital in Year before AMI (2)	Report Cards × Prior Year Admission (3)	Effect of Report Cards (4)	Admission to Hospital in Year before AMI (5)	Report Cards × Prior Year Admission (6)
CABG within one year of admission (1=yes)	.60** (.21)			.91** (.44)		
[14.76, 9.10]	.81** (.15)	-3.80** (.15)	-.65 (.44)	1.39** (.42)	-3.78** (.16)	-1.52** (.19)
CABG within one day of admission (1=yes)	-.78** (.29)			-.59** (.23)		
[5.40, 2.97]	-.97** (.40)	-1.73** (.13)	.72* (.41)	-.66** (.30)	-1.71** (.14)	.39 (.30)
PTCA within one year of admission (1=yes)	-1.69 (1.22)			-1.22 (1.17)		
[13.94, 8.76]	-1.73 (1.55)	-3.50** (.17)	.23 (1.15)	-.96 (1.46)	-3.46** (.19)	-.76 (.99)

Table 4, part 1

PTCA within one day of admission (1=yes)	-2.21** (.85)			-2.06** (.91)		
[7.81, 4.82]	-2.55** (1.05)	-2.05** (.16)	1.22* (.70)	-2.92** (1.07)	-2.00** (.18)	.59 (.57)
Cath within one year of admission (1=yes)	-.81 (1.02)			.24 (.56)		
[40.65, 26.77]	-.88 (1.48)	-9.55** (.34)	.48 (1.64)	.72 (.89)	-9.47** (.38)	-1.37 (1.16)
Cath within one day of admission (1=yes)	-3.73** (1.51)			-2.77** (1.17)		
[26.81, 16.25]	-4.28** (1.90)	-7.54** (.38)	2.02 (1.40)	-2.86* (1.46)	-7.45** (.41)	.56 (1.08)

NOTE.—Standard errors are based on an estimator of the variance-covariance matrix that is consistent in the presence of heteroscedasticity and of any correlation of regression errors within states over time. Coefficients and standard errors are multiplied by 100 to facilitate interpretation. For expenditure models, N = 1,770,585; for all other models, N = 1,770,452.

* Numbers in brackets are the means for individuals without and with a prior year hospital admission.

** Significantly different from zero at the 10 percent level.

** Significantly different from zero at the 5 percent level.

Table 4, part 2

7.4 Table 5: expenditures and health outcomes for AMI patients

Read:

- Report cards increase total hospital expenditures after AMI admission.
- They increase readmission with heart failure.
- There is evidence of increased mortality, especially for sicker patients with prior admissions.
- The interaction terms show that the adverse effects are concentrated among high-severity patients.

Economic message: The welfare test is negative: report cards raised costs and worsened some health outcomes rather than improving health at lower cost.

TABLE 5
EFFECTS OF REPORT CARDS ON HOSPITAL EXPENDITURES AND HEALTH OUTCOMES: MEDICARE BENEFICIARIES WITH AMI, 1987–94

DEPENDENT VARIABLE	A. ASSUMES REPORT CARDS EFFECTIVE 1991 IN N.Y. AND 1993 IN PA.			B. ASSUMES REPORT CARDS EFFECTIVE 1993 IN N.Y. AND PA.		
	Effect of Report Cards (1)	Admission to Hospital in Year before AMI (2)	Report Cards × Prior Year Admission (3)	Effect of Report Cards (4)	Admission to Hospital in Year before AMI (5)	Report Cards × Prior Year Admission (6)
ln(total hospital expenditures in year after admission)	3.92** (1.08)			3.95** (1.52)		
	2.89** (.73)	7.33** (.48)	3.35* (1.75)	3.31** (1.16)	7.44** (.53)	1.93 (1.49)
Readmission with AMI within one year of admission (1=yes)	.02 (.08)			.06 (.07)		
	-.15 (.10)	1.70** (.06)	.55** (.13)	-.11 (.09)	1.72** (.06)	.52** (.14)
Readmission with heart failure within one year of admission (1=yes)	.50** (.10)			.54** (.10)		
	-.20** (.08)	4.89** (.10)	2.27** (.26)	-.18** (.08)	4.93** (.11)	2.30** (.36)
Mortality within one year of admission (1=yes)	.45 (.32)			.45* (.26)		
	.37 (.41)	11.90** (.09)	-.02 (.44)	.13 (.27)	11.88** (.10)	.69** (.13)

NOTE.—Standard errors are based on an estimator of the variance-covariance matrix that is consistent in the presence of heteroscedasticity and of any correlation of regression errors within states over time. Coefficients and standard errors are multiplied by 100 to facilitate interpretation. For expenditure models, $N = 1,708,585$; for all other models, $N = 1,770,452$.

* Significantly different from zero at the 10 percent level.

** Coefficient different from zero at the 5 percent level.

7.5 Table 6: alternative model specifications

Read:

- Table 6 tests alternative modelling assumptions.
- Panel A restricts comparison states to New York/Pennsylvania and neighbouring states.
- Panel B adds linear time trends for New York and Pennsylvania.
- The core conclusions on treatment delay, expenditures, and adverse outcomes are robust.

Economic message: The main findings are not driven by using all non-report-card states as controls or by simple differential trends in the report-card states.

TABLE 6
ALTERNATIVE MODELS OF EFFECTS OF REPORT CARDS ON CABG SURGERY RATES, HOSPITAL EXPENDITURES, AND HEALTH OUTCOMES
OF INDIVIDUAL MEDICARE BENEFICIARIES WITH AMI, 1987–94

DEPENDENT VARIABLE	A. HOSPITALS AND PATIENTS FROM N.Y., PA., CONN., MD., AND N.J. ONLY			B. LINEAR TIME TREND INCLUDED FOR N.Y. AND PA.		
	Effect of Report Cards (1)	Admission to Hospital in Year before AMI (2)	Report Cards × Prior Year Admission (3)	Effect of Report Cards (4)	Admission to Hospital in Year before AMI (5)	Report Cards × Prior Year Admission (6)
CABG within one year of admission (1=yes)	.42 (.36)			.27** (.10)		
	.08** (.31)	-2.66** (.41)	-1.94** (.22)	.46** (.18)	-3.80** (.15)	-.65 (.44)
PTCA within one year of admission (1=yes)	-.05 (.06)			.10 (.05)		
	-.50 (1.24)	-2.11** (.44)	-1.50 (.90)	.03 (1.00)	-3.50** (.17)	.23 (1.15)
Cath within one day of admission (1=yes)	-.76 (1.82)			.70 (1.23)		
	-.22 (2.12)	-4.63** (.61)	-1.87 (1.15)	.10 (1.62)	-7.54** (.38)	2.01 (1.40)

Table 6, part 1

ln(total hospital expenditures in year after admission)	1.74 (1.19)			3.49 (2.81)		
	1.96 (1.40)	10.81** (.97)	-.62 (1.83)	2.52 (3.25)	7.33** (.48)	3.36* (1.75)
Readmission with AMI within one year of admission (1=yes)	.17 (.09)			-.23** (.07)		
	.07 (.11)	1.90** (.12)	.35 (.21)	-.38** (.07)	1.70** (.06)	.55** (.15)
Readmission with heart failure within one year of admission (1=yes)	.41** (.09)			.01 (.10)		
	-.04 (.12)	5.52** (.15)	1.57** (.36)	-.64** (.12)	4.80** (.10)	2.27** (.26)
Mortality within one year of admission (1=yes)	.44* (.16)			.10 (.30)		
	.51 (.25)	12.05** (.23)	-.11 (.41)	.13 (.45)	11.90** (.09)	-.02 (.44)

NOTE.—Models assume that report cards are effective 1991 in New York and 1993 in Pennsylvania. Standard errors are based on an estimator of the variance-covariance matrix that is consistent in the presence of heteroscedasticity and of any correlation of regression errors within states over time. Coefficients and standard errors are multiplied by 100 to facilitate interpretation. For expenditure models in panel A, $N = 360,925$; for all other models in panel A, $N = 307,421$. For expenditure models in panel B, $N = 1,708,585$; for all other models in panel B, $N = 1,770,452$.

* Significantly different from zero at the 10 percent level.

Table 6, part 2

7.6 Table 7: CABG-patient-only analysis

Read:

- Table 7 repeats outcome regressions on the population of CABG recipients, similar to prior literature.
- The CABG-only analysis suggests more favorable effects, including lower mortality among CABG patients.
- This contrasts with the AMI at-risk population results.

Economic message: Conditioning on receiving CABG can make report cards look beneficial because the CABG patient pool changes. This is the central selection lesson.

TABLE 7
EFFECTS OF REPORT CARDS ON TOTAL HOSPITAL EXPENDITURES AND HEALTH OUTCOMES OF INDIVIDUAL MEDICARE BENEFICIARIES RECEIVING CABG SURGERY, 1987–94

DEPENDENT VARIABLE	A. ASSUMES REPORT CARDS EFFECTIVE 1991 IN N.Y. AND 1993 IN PA.			B. ASSUMES REPORT CARDS EFFECTIVE 1993 IN N.Y. AND PA.		
	Effect of Report Cards (1)	Admission to Hospital in Year before AMI (2)	Report Cards × Prior Year Admission (3)	Effect of Report Cards (4)	Admission to Hospital in Year before AMI (5)	Report Cards × Prior Year Admission (6)
ln(total hospital expenditures in year after admission)	8.28** (3.30)			5.93** (2.67)		
	7.08** (3.42)	2.48** (.39)	2.72** (.80)	4.78** (2.68)	2.52** (.37)	2.80** (.73)
Readmission with AMI within one year of admission (1=yes)	-.10* (.05)			-.17** (.04)		
	-.15** (.04)	.22** (.02)	.10 (.09)	-.17** (.06)	.23** (.02)	.00 (.09)
Readmission with heart failure within one year of admission (1=yes)	.14 (.18)			.31** (.14)		
	-.01 (.11)	3.47** (.07)	.42 (.49)	.02 (.14)	3.46** (.07)	.86* (.46)
Mortality within one year of admission (1=yes)	-1.17** (.28)			-1.02** (.38)		
	-1.02** (.32)	2.72** (.16)	-.24 (.23)	-.86* (.46)	2.72** (.16)	-.21 (.25)

NOTE.—Standard errors are based on an estimator of the variance-covariance matrix that is consistent in the presence of heteroscedasticity and of any correlation of regression errors within states over time. Coefficients and standard errors are multiplied by 100 to facilitate interpretation. For expenditure models, $N = 967,882$; for all other models, $N = 965,942$.
* Significantly different from zero at the 10 percent level.
** Significantly different from zero at the 5 percent level.

7.7 Appendix tables: descriptive statistics

Read:

- Appendix Table A1 reports hospital-level descriptive statistics, including patient severity variation, hospital size, teaching status, ownership, and other hospital characteristics.
- Appendix Table A2 reports patient-level descriptive statistics for AMI and CABG cohorts in 1987 and 1994.
- These tables document the baseline severity distributions and treatment rates underlying the empirical analysis.

Economic message: The appendix shows that CABG and AMI patients differ substantially and that patient severity is highly skewed, which makes selection behaviour economically important.

ards on the performance of schools raise the same issues and therefore also need careful empirical evaluation.

ppendix

TABLE A1
DESCRIPTIVE STATISTICS ON HOSPITALS

	WEIGHTED BY AND USING HEALTH HISTORIES OF AMI PATIENTS		WEIGHTED BY AND USING HEALTH HISTORIES OF CABG PATIENTS	
	1987	1994	1987	1994
N of patients' total hospital expenditures one year prior to admission	2,199 (.445)	2,166 (.587)	1,556 (.351)	1,934 (.281)
N of patients' total days in hospital one year prior to admission	2,439 (.574)	2,473 (.751)	1,699 (.294)	2,245 (.418)
Number of hospitals in the state	180.3	157.7	31.58	36.52
Hospital size medium (1=yes)	49.7%	51.9%	35.8%	46.5%
Hospital size large	25.3%	20.9%	63.8%	51.0%
Teaching hospital	19.1%	20.5%	46.2%	44.1%
Public ownership	15.7%	13.3%	10.1%	8.7%
For-profit ownership	10.4%	10.1%	7.5%	8.4%
Urban location	26.4%	24.5%	2.7%	3.8%
Subject to report cards	.00	14.2%	.00	13.5%
Sample size	5,369	4,792	739	936

Appendix Table A1: hospital statistics

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TABLE A2
DESCRIPTIVE STATISTICS ON ELDERLY MEDICARE BENEFICIARIES WITH AMI AND ELDERLY MEDICARE BENEFICIARIES RECEIVING CABG SURGERY

	WITH AMI		RECEIVING CABG SURGERY	
	1987	1994	1987	1994
Total hospital expenditures one year prior to admission	\$2,690 (6,493)	\$2,977 (7,464)	\$4,431 (7,188)	\$3,771 (7,586)
Total days in hospital one year prior to admission	4.21 (11.48)	4.22 (13.48)	4.97 (8.63)	3.39 (8.05)
Total hospital expenditures one year after admission	\$14,634 (13,381)	\$18,959 (19,060)	\$39,226 (13,857)	\$34,474 (22,460)
CABG within one year of admission (1=yes)	9.2%	16.2%	100%	100%
Readmission with AMI within one year of admission	5.8%	5.5%	1.1%	1.2%
Readmission with heart failure within one year of admission	9.0%	9.4%	6.1%	6.6%
Mortality within one year of admission	40.2%	32.9%	12.2%	10.7%
Age	76.0	76.4%	71.39	72.54
Gender (1=female)	49.8%	48.7%	34.2%	34.7%
Race (1=black)	5.5%	5.9%	2.4%	3.4%

Appendix Table A2: patient statistics

8 Detailed result synthesis

8.1 Selection effects

Report cards decreased the average severity of patients receiving CABG in New York and Pennsylvania relative to comparison states. The most direct evidence is that prior-year expenditures for CABG recipients fell in treatment states while the overall AMI population did not become healthier. This is consistent with providers avoiding high-risk CABG candidates.

8.2 Matching effects

Report cards also improved matching by directing sicker patients toward teaching hospitals. This is the positive channel of report cards: better information can help high-risk patients find more capable providers. However, the welfare results show that this beneficial channel was not strong enough to offset selection and treatment distortions.

8.3 Quantity and timing effects

Report cards increased CABG use among healthier patients but delayed rapid intervention. One-day CABG, PTCA, and catheterization rates fall. This suggests providers became more cautious or selective about intensive treatment after report cards.

8.4 Welfare effects

Hospital expenditures increased, readmissions with heart failure increased, and mortality worsened for sicker patients. These effects imply that report cards decreased patient and social welfare in the short run.

9 Short summary

- The paper studies CABG report cards in New York and Pennsylvania.
- Report cards were meant to improve information about provider quality.
- The policy also gave providers incentives to avoid high-risk patients.
- The paper studies all AMI patients, not only CABG recipients, to avoid conditioning on treatment selection.
- Report cards made the CABG patient pool healthier, consistent with provider selection.
- Sicker CABG patients shifted toward teaching hospitals, consistent with improved matching.
- CABG use rose for healthier AMI patients, while PTCA and rapid treatment declined.
- Hospital expenditures increased.
- Readmissions with heart failure increased, and mortality worsened for sick patients.
- CABG-only analyses can incorrectly suggest quality improvements because report cards change who receives CABG.
- The overall short-run welfare effect is negative.

10 Conclusion

10.1 Core conclusion

Dranove, Kessler, McClellan, and Satterthwaite show that report cards can backfire in health care markets. Although they provide information about provider quality, they also alter provider incentives. In the CABG

setting, report cards induced selection away from severely ill patients and worsened outcomes for the broader at-risk AMI population.

10.2 Economic intuition

When providers are evaluated on risk-adjusted outcomes, they care about the observable composition of treated patients. If risk adjustment is imperfect, providers can improve measured performance by avoiding patients whose unobserved severity makes bad outcomes more likely. This can improve reported mortality among treated patients while harming the broader population at risk.

10.3 Incremental takeaway

The paper's lasting contribution is to show that information policy must be evaluated in equilibrium. Public performance measures change patient sorting, provider selection, treatment intensity, and welfare. More information is not automatically better when measured agents can change the population on which they are measured.